Student ID: ____

Full Names: ____

Full-stack Enterprise Software Developer Test

(May 2026)

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1. The time allotted for completing this test is 2 hours (10:30 – 12:30).
2. You are expected to use your Computer with an IDE or any Code Editor tool of your choice to implement your solution for the question.
3. *For the tasks in the question, you are expected to take screenshot(s) of your result(s), save each into a .png or .jpg image file, placed inside a folder named, screenshots and include these in your submission, making sure to include all your project source code, pushed to a repository on your Github account. For the given question, when you have completed your own solution, you are required to take each of the set of 5 evidential sample screenshots, which have been included at the end of the question.*
4. Upon completion, to submit your work for review and grading, simply push your entire Project Source Code folder (including the screenshots folder) to the repository on Github.
5. **This Career Center test belongs to MIU CS Department and must not be copied or photographed or reproduced or transferred or shared or distributed. Any violation will be penalized.**

Make sure to include the screenshots of your results, as required.

Software Development Tasks (100 points)

Test of Software Design, Problem-solving, Development and Coding skills

Note: *For the tasks in this question, you are expected to take screenshot(s) of your result(s), save each into a .png or .jpg image file, placed inside a folder named, screenshots and include these in the project folder, which you commit to git and submit. From your own solution, you are required to take each of the set of 5 or more evidential sample screenshots, which have been included at the end of the question.*

1. (100 points) **Implement an end-to-end, full-stack data-driven enterprise web application**

Assume you have been hired by a bank named, MIU Bank of Iowa. And they want you to design and develop a basic web-based software solution, which they call, "Customer-Accounts Management system" (CAMS), which they will be using to manage their customer-accounts data.

Especially important to the Bank Manager is, the current Liquidity Position of the bank, which is calculated by summing the total balance of all the customer-accounts. Also important to the Bank Manager, is the **Prime Accounts. A Prime Account** is account whose balance is greater than $50,000.

For the purpose of this test, here is a description of the solution model for the system:

A Customer can own one or many Account(s).
And, an Account can be owned by one or multiple Customer(s).

Your solution model should consist of the following two data entity:

1. Customer
2. Account

Here are the attributes for the entities, including some useful descriptions and/or sample data values:

Customer:

customerId: long, (Primary Key field)

firstName, (required field) (e.g. Bob, Anna, Carlos etc.)

lastName, (required field) (e.g. Jones, Smith etc.)

Account:

accountId: long, (Primary Key field)

accountNumber, (required field; unique) (e.g. AC1001, AC1002, etc.)

accountType, (required field) (e.g. Checking, Savings)

dateOpened, (optional field) (e.g. 2021-12-3, 2022-5-21, etc.)

balance, (required field) – This is the amount of money (in dollars and cents) in the account

Data:

Here is the Bank's existing sample data, which you are expected to input into your database:

Customer data:

Customer Id	First Name	Last Name
1	Bob	Jones
2	Anna	Smith
3	Carlos	Jimenez

Account data:

Account Id	Account Number	Account Type	Date Opened	Balance	Customer Id
1	AC1002	Checking	2022-07-10	55,900.50	1, 2
2	AC1001	Savings	2021-11-15	125.95	1
3	AC1003	Savings	2022-07-11	75,000	3

For this question, you are required to do the following:

1. Draw a Domain model UML Class diagram for the solution, showing only the two (2) entity classes – Customer and Account.

2. Using your preferred set of tools, technologies and frameworks, etc., (or some other Enterprise Web application development language/platform/tool(s) that you prefer), implement a working web application for the Bank. You may use any database of your choice.

You are expected to implement only the following features and use-cases:

1. Display a homepage which presents a set of menu options. (see sample screenshot below)

2. Display list of all Accounts (Allows the user to view a list of all the accounts registered in the system, including the customer data). The Bank requires this list to be displayed sorted in descending order of the Account balance amounts.

   Also, display at the bottom of this list, the Liquidity Position of the bank, which is the sum of all Account balances. (see sample screenshot below)

3. Implement a Web Form UI and backend functionality for creating a new Customer-Account.

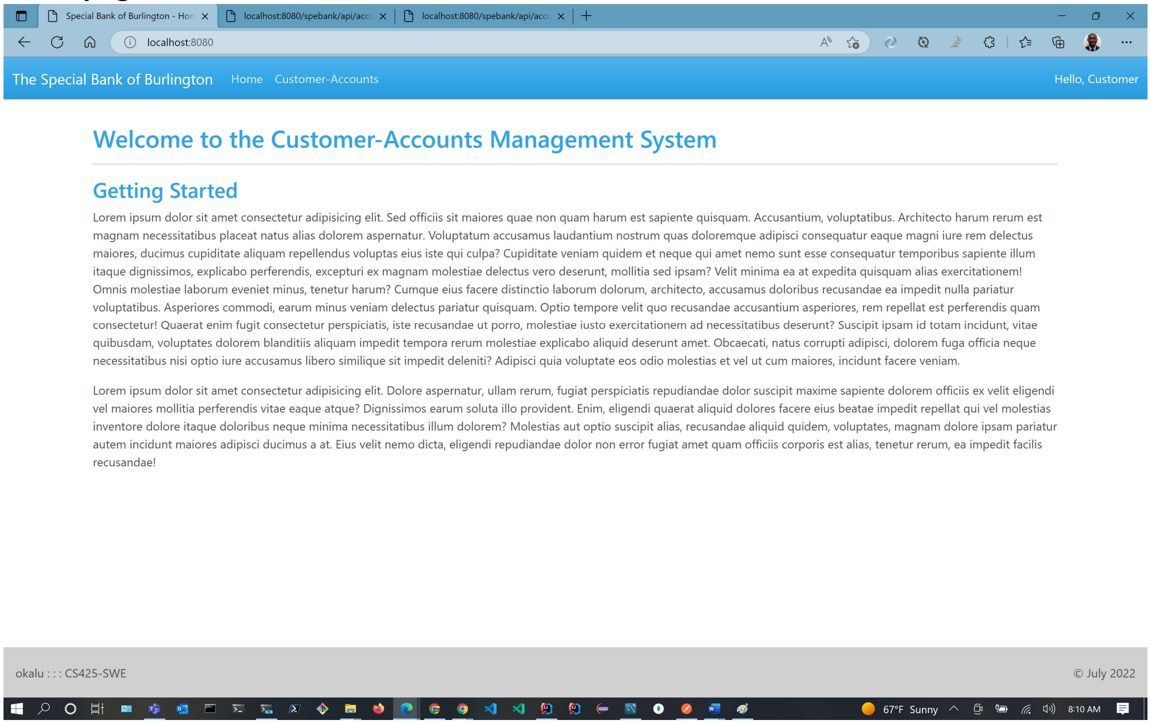
4. Implement a RESTful Web Service Endpoint (i.e. Web API) url which presents the list of all Customers and their Account(s) data, in JSON format.

5. Implement a RESTful Web Service Endpoint (i.e. Web API) url which presents the list of only Prime Accounts in JSON format. Note: Prime Accounts are accounts that have more than $50,000.00.

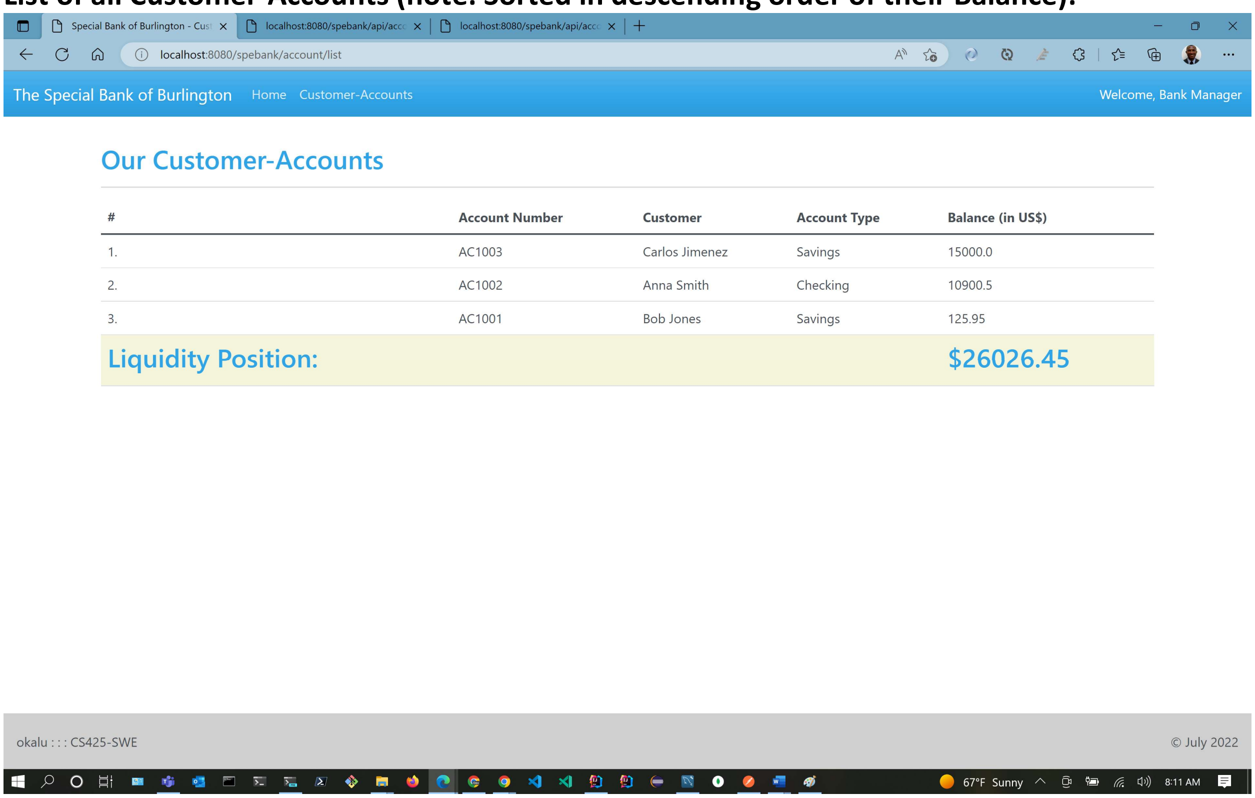
Shown below are sample User Interfaces and data presentation for the above features/use-cases.

Note: Your own UI design does NOT necessarily have to look exactly like these samples. But your UIs should contain/present all the necessary data and data fields, as required.

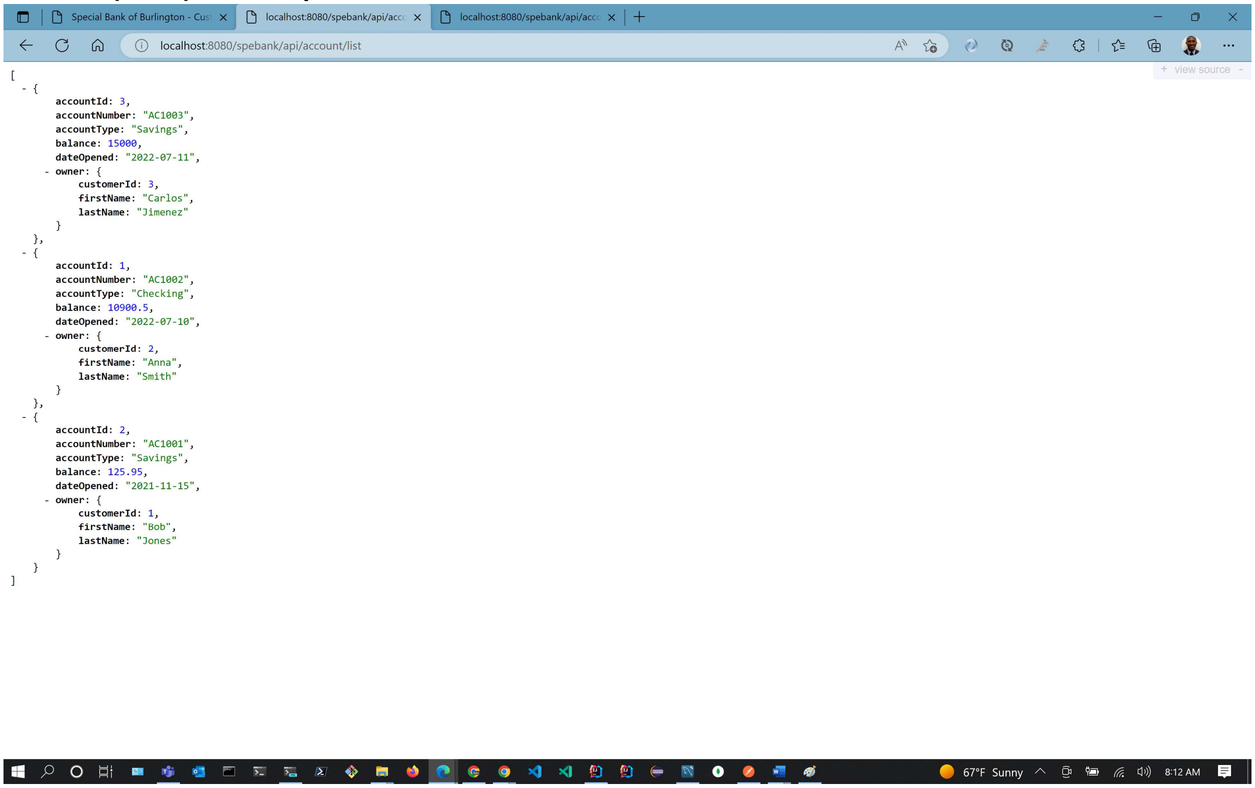
Homepage:



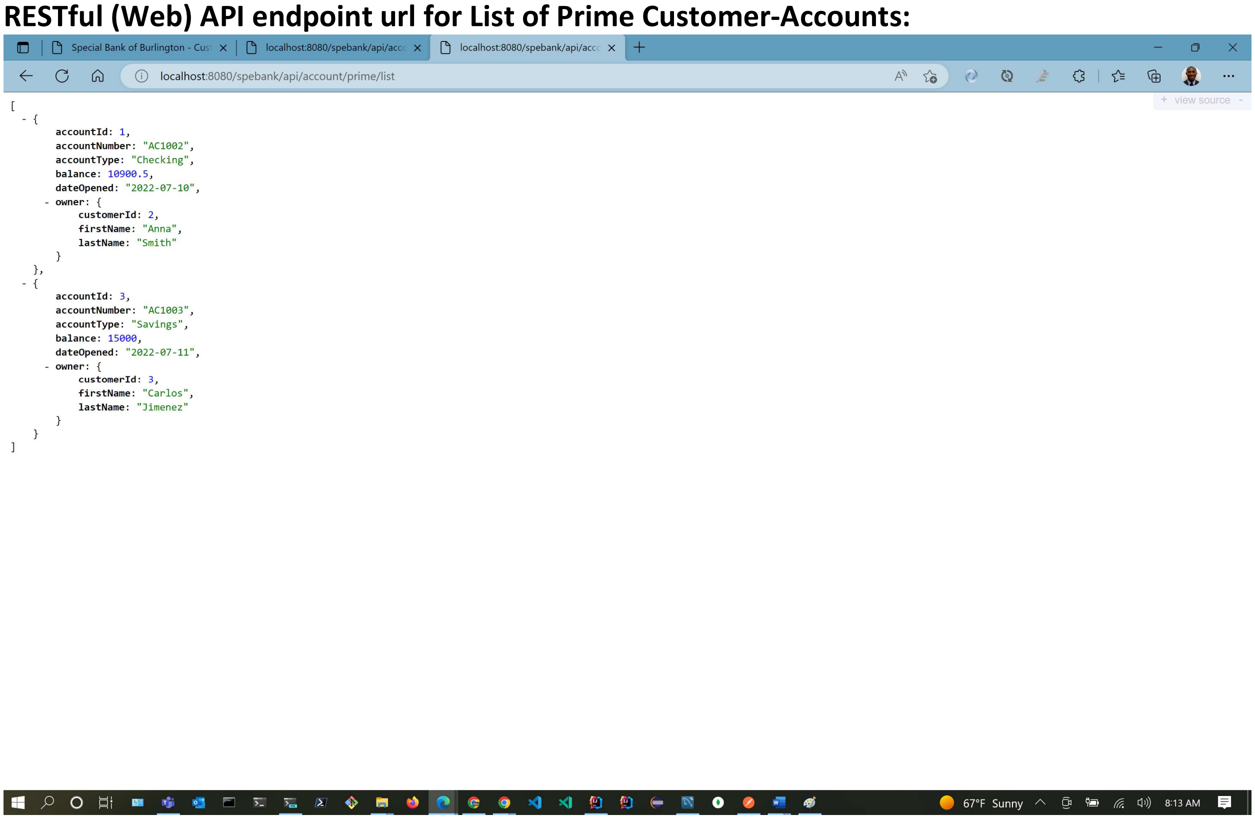
List of all Customer-Accounts (note: Sorted in descending order of their Balance):



RESTful (Web) API endpoint url for List of all Customer-Accounts:

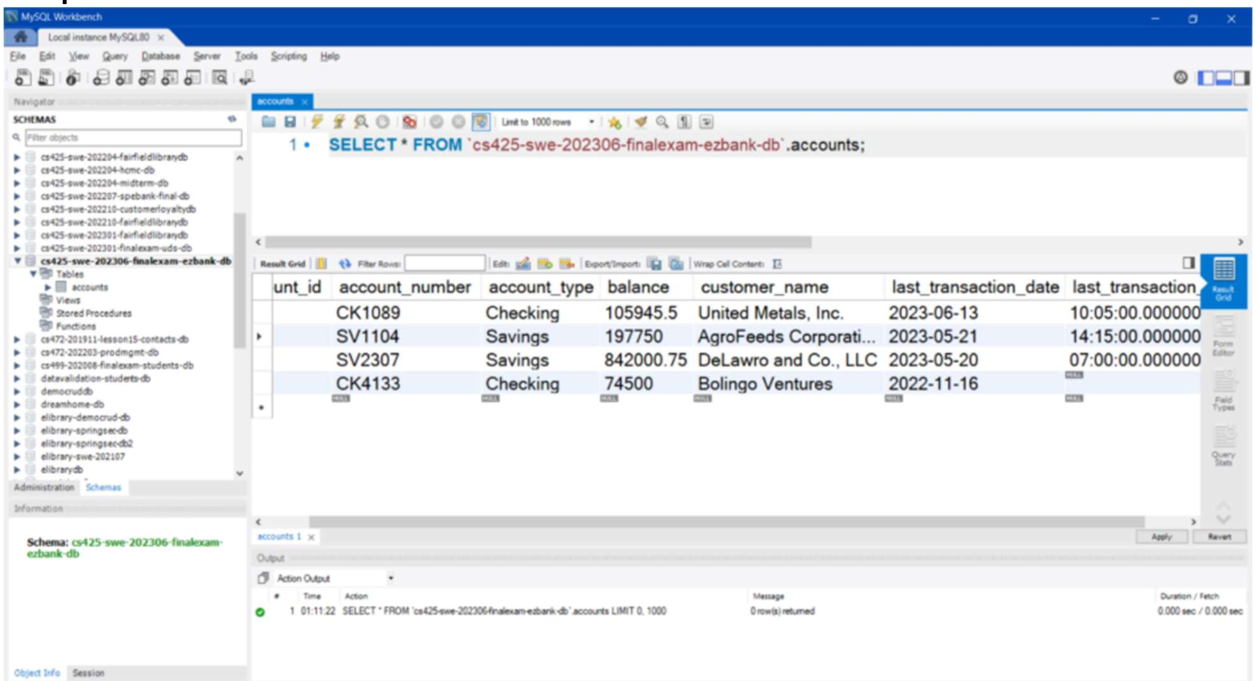


RESTful (Web) API endpoint url for List of Prime Customer-Accounts:



Database Tables screenshot (take a screenshot of your database table(s), showing the data, and similar to the one pasted below):

Sample DB table



//-- The End --//